

Basel Framework

https://www.bis.org/basel_framework/

<https://www.bis.org/baselframework/BaselFramework.pdf>

The Basel Framework is a consolidated version of the full set of standards of the Basel Committee on Banking Supervision (BCBS), which is the primary global standard setter for the prudential regulation of banks. The membership of the BCBS has agreed to fully implement these standards and apply them to the internationally active banks in their jurisdictions.

The framework comprises the 14 standards listed below. Each standard is divided into chapters, and many chapters have multiple versions, eg a chapter may have a version that is applicable now and one that will become applicable after the Basel III reforms have been implemented.

Acronym	Standard names
SCO	Scope and definitions
CAP	Definition of capital
RBC	Risk-based capital requirements
CRE	Calculation of RWA for credit risk
MAR	Calculation of RWA for market risk
OPE	Calculation of RWA for operational risk
LEV	Leverage ratio
LCR	Liquidity Coverage Ratio
NSF	Net Stable Funding Ratio
LEX	Large exposures
MGN	Margin requirements
SRP	Supervisory review process
DIS	Disclosure requirements
BCP	Core Principles for Effective Banking Supervision

CAP

This standard describes the criteria that bank capital instruments must meet to be eligible to satisfy the Basel capital requirements, as well as necessary regulatory adjustments and transitional arrangements.

CAP10: Eligible Capital

This chapter sets out the eligibility criteria for regulatory capital. Three categories of instruments are permitted: Common Equity Tier 1, Additional Tier 1 and Tier 2.

Capital Components

10.1 Regulatory capital consists of three categories, each governed by a single set of criteria that instruments are required to meet before inclusion in the relevant category.

- Common Equity Tier 1 (going-concern capital)
- Additional Tier 1 (going-concern capital)
- Tier 2 Capital (gone-concern capital)

10.2 Total regulatory capital is the sum of Common Equity Tier 1, Additional Tier 1 and Tier 2 capital, net of regulatory adjustments described in CAP30. Tier 1 capital is the sum of Common Equity Tier 1 and Additional Tier 1 capital, net of the regulatory adjustments in CAP30 applied to those categories.

10.3 It is critical that banks' risk exposures are backed by a high-quality capital base. To this end, the predominant form of Tier 1 capital must be common shares and retained earnings.

Table 1.11 Comparison between Basel II and Basel III regulatory capital treatment of some (major) capital components

Capital category	Basel II	Basel III
Common shares	Tier 1 – common equity	Tier 1 – common equity
Retained earnings	Tier 1 – common equity	Tier 1 – common equity
Innovative capital instruments	Tier 1 – additional	Excluded and grandfathered
Non-innovative capital instruments	Tier 1 – additional	Included in Tier 1 – additional, subject to conditions
Subordinate debt	Tier 2	Included in Tier 2, subject to conditions

Common Equity Tier 1

10.6 Common Equity Tier 1 capital consists of the sum of the following elements:

- Common shares issued by the bank that meet the criteria for classification as common shares for regulatory purposes (or the equivalent for non-joint stock companies);
- Stock surplus (share premium) resulting from the issue of instruments included Common Equity Tier 1;
- Retained earnings;
- Accumulated other comprehensive income and other disclosed reserves; and
- Regulatory adjustments applied in the calculation of Common Equity Tier 1.

10.7 Dividends are removed from Common Equity Tier 1 in accordance with applicable accounting standards.

Additional Tier 1

10.9 Additional Tier 1 capital consists of the sum of the following elements:

- instruments issued by the bank that meet the criteria for inclusion in Additional Tier 1 capital (and are not included in Common Equity Tier 1);
- stock surplus (share premium) resulting from the issue of instruments included in Additional Tier 1 capital;

10.11 The following criteria must be met or exceeded for an instrument issued by the bank to be included in Additional Tier 1 capital.

- Subordinated to depositors, general creditors and subordinated debt of the bank. In the case of an issue by a holding company, the instrument must be subordinated to all general creditors.
- Is neither secured nor covered by a guarantee of the issuer or related entity or other arrangement that legally or economically enhances the seniority of the claim vis-à-vis bank creditors
- Is perpetual, ie there is no maturity date and there are no step-ups or other incentives to redeem

Tier 2

10.14 Tier 2 capital consists of the sum of the following elements:

- instruments issued by the bank that meet the criteria for inclusion in Tier 2 capital (and are not included in Tier 1 capital);
- stock surplus (share premium) resulting from the issue of instruments included in Tier 2 capital;
- certain loan-loss provisions as specified in CAP10.18 and CAP10.19;

10.16 The objective of Tier 2 is to provide loss absorption on a gone-concern basis. Based on this objective, the following criteria must be met or exceeded for an instrument to be included in Tier 2 capital.

- Subordinated to depositors and general creditors of the bank. In the case of an issue by a holding company, the instrument must be subordinated to all general creditors.
- Is neither secured nor covered by a guarantee of the issuer or related entity or other arrangement that legally or economically enhances the seniority of the claim vis-à-vis depositors and general bank creditors
- Maturity:
 - Minimum original maturity of at least five years
 - Recognition in regulatory capital in the remaining five years before maturity will be amortised on a straight line basis.

10.18 Under the standardised approach to credit risk, provisions or loan-loss reserves held against future, presently unidentified losses are freely available to meet losses which subsequently materialise and therefore qualify for inclusion within Tier 2. Provisions ascribed to identified deterioration of particular assets or known liabilities, whether individual or grouped, should be excluded. Furthermore, general provisions/general loan-loss reserves eligible for inclusion in Tier 2, measured gross of tax effects, will be limited to a maximum of 1.25% of credit risk-weighted assets (RWA) calculated under the standardised approach.

10.19 Under the internal ratings based approach, where the total expected loss amount is less than total eligible provisions (measured gross of tax effects), as explained in CRE35, banks may recognise the difference in Tier 2 capital up to a maximum of 0.6% of credit risk-weighted assets calculated under the internal ratings-based approach. At national discretion, a limit lower than 0.6% may be applied.

CAP30: Adjustments

This chapter describes adjustments that must be made to the components of regulatory capital in order to calculate the amount of a bank's capital resources that may be used to meet prudential requirements.

Minimum TLAC

30.2 Global systemically important banks (G-SIBs) are required to meet a minimum total loss-absorbing capacity (TLAC) requirement set in accordance with the Financial Stability Board's (FSB) TLAC principles and

term sheet. The criteria for an instrument to be recognised as TLAC by the issuing G-SIB are set out in the FSB's TLAC Term Sheet. Bank that invest in TLAC or similar instruments may be required to deduct them in the calculation of their own regulatory capital.

<https://www.fsb.org/uploads/TLAC-Principles-and-Term-Sheet-for-publication-final.pdf>

4. Minimum TLAC must be at least 16% of the resolution group's RWAs ("TLAC RWA Minimum") as from 1 January 2019 and at least 18% as from 1 January 2022. This requirement does not include any applicable regulatory capital (Basel III) buffers, which must be met in addition to the TLAC RWA Minimum.

5. Minimum TLAC is an additional requirement to minimum regulatory capital requirements.

Goodwill & Other Intangibles

30.7 Goodwill and all other intangibles must be deducted in the calculation of Common Equity Tier 1, including any goodwill included in the valuation of significant investments in the capital of banking, financial and insurance entities that are outside the scope of regulatory consolidation.

Deferred Tax Assets

30.9 Deferred tax assets (DTAs) that rely on future profitability of the bank to be realised are to be deducted in the calculation of Common Equity Tier 1. DTAs may be netted with associated DTLs only if the DTAs and DTLs relate to taxes levied by the same taxation authority and offsetting is permitted by the relevant taxation authority

CAP50: Prudent Valuation Guidance

This chapter provides banks with guidance on prudent valuation for positions that are accounted for at fair value, whether they are in the trading book or in the banking book.

Marking-to-Market

50.4 Marking-to-market is at least the daily valuation of positions at readily available close out prices that are sourced independently. Examples of readily available close out prices include exchange prices, screen prices, or quotes from several independent reputable brokers.

50.5 Banks must mark-to-market as much as possible.

Marking-to-Model

50.6 Only where marking-to-market is not possible should banks mark-to-model, but this must be demonstrated to be prudent. Marking-to-model is defined as any valuation which has to be benchmarked, extrapolated or otherwise calculated from a market input. When marking to model, an extra degree of conservatism is appropriate. Valuation adjustments should be made as appropriate, for example, to cover the uncertainty of the model valuation.

RBC

This standard describes the framework for risk-based capital requirements.

RBC20: MCR

This chapter sets out the minimum regulatory capital requirements under the risk-based framework and how banks must calculate risk-weighted assets.

MCR

20.1 Banks must meet the following requirements at all times:

- Common Equity Tier 1 must be at least 4.5% of risk-weighted assets (RWA).
- Tier 1 capital must be at least 6% of RWA.
- Total capital must be at least 8% of RWA.

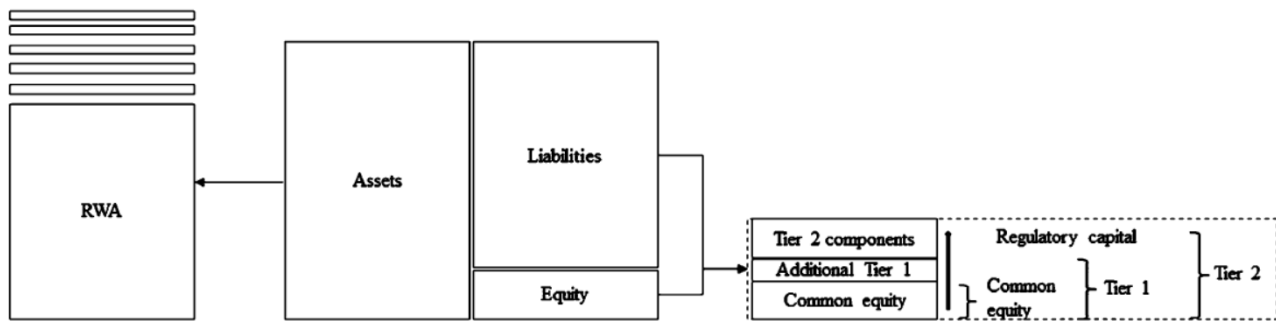
Common Equity Tier 1 (CET1) Ratio	Tier 1 Ratio	Total Capital Ratio
(Common Equity –Deductions) RWA	(CE + AT1 - Deductions) RWA	(Tier 1 + Tier 2 - Deductions) RWA
▪ Common Equity includes common shares, retained earnings, reserves. ▪ Deductions includes goodwill, other intangibles, treasury shares, excess minority interests, deferred tax assets, mortgage servicing rights, investments in financial institutions over 10% of share capital.	▪ AT 1 instruments (such as hybrid capital instruments) are required to have enhanced loss absorbency features in addition to coupon deferability.	▪ Tier 2 includes subordinated debt instruments with enhanced loss absorbency features, e.g. permanent principal write-down or conversion into equity at ‘point of non-viability’ of issuing bank.

RWA

20.3 The Basel framework describes how to calculate RWA for credit risk, market risk and operational risk. The requirements for calculating RWA for credit risk and market risk allow banks to use different approaches, some of which banks may only use with supervisory approval.

20.4 The RWA that banks must use to determine compliance with the requirements set out in RBC20.1 (and the buffers in RBC30 and RBC40) is the higher of:

1. the sum of the following three elements, calculated using the bank’s nominated approaches:
 - RWA for credit risk (as calculated in RBC20.6 to RBC20.8);
 - RWA for market risk (as calculated in RBC20.9); and
 - RWA for operational risk (as calculated in RBC20.10); and
2. 72.5% of the sum of the elements listed in point (1) above, calculated using only the standardised approaches listed in RBC20.11. This element of this requirement is referred to as the output floor, and the RWA amount that is multiplied by 72.5% is referred to as the base of the output floor.



Credit Risk RWA

20.6 RWA for credit risk (including counterparty credit risk) is calculated as the sum of the following:

- 1. Credit RWA for banking book exposures, except the RWA listed in (2) to (6) below, calculated using:
 - The standardised approach, set out in CRE20 to CRE22; or
 - The internal ratings-based (IRB) approach, set out in CRE30 to CRE36.
- 2. RWA for counterparty credit risk arising from banking book exposures and from trading book instruments (as specified in CRE55), except the exposures listed in (3) to (6) below, using the methods outlined in CRE51.

Market Risk RWA

20.9 RWA for market risk is calculated as the sum of the following:

- 1. RWA for market risk for instruments in the trading book and for foreign exchange risk and commodities risk for exposures in the banking book, calculated using one or more of the following approaches:
 - The standardised approach for market risk, set out in MAR20 to MAR23;
 - The internal models approach (IMA) for market risk, set out in MAR30 to MAR33; or
 - The simplified standardised approach for market risk, set out in MAR40.
- 2. RWA for credit valuation adjustment (CVA) risk in the banking and trading book, calculated using one of the following methods set out in MAR50:
 - The basic approach to CVA risk (BA-CVA).
 - The standardised approach to CVA risk (SA-CVA).
 - 100% of the bank’s RWA for counterparty credit risk, for banks that have exposures below a materiality threshold (see MAR50.9).

Operational Risk RWA

20.10 RWA for operational risk is calculated using the standardised approach for operational risk, set out in OPE25.

RBC25: Banking vs Trading Book

This chapter sets out the instruments to be included in the trading book (which are subject to market risk capital requirements) and those to be included in the banking book (which are subject to credit risk capital requirements).

Banking Book

25.8 The following instruments must be assigned to the banking book:

- retail and small or medium-sized enterprise (SME) credit

Trading Book

25.9 There is a general presumption that any of the following instruments are being held for at least one of the purposes listed in RBC25.5 and therefore are trading book instruments, unless specifically otherwise provided for in RBC25.3 or RBC25.

- instruments held as accounting trading assets or liabilities;
- listed equities;
- trading-related repo-style transaction; or
- options including embedded derivatives

Under IFRS (IAS 39) and US GAAP, these instruments would be designated as held for trading. Under IFRS 9, these instruments would be held within a trading business model. These instruments would be fair valued through the P&L account

RBC30: Buffers

This chapter describes buffers that banks are expected to maintain above the minimum riskbased capital requirements, as well as the capital conservation requirements that apply to banks that do not maintain such buffers.

Capital Conservation Buffer (CCoB)

30.1 This chapter outlines the operation of the capital conservation buffer, which is designed to ensure that banks build up capital buffers outside periods of stress which can be drawn down as losses are incurred. The requirement is based on simple capital conservation rules designed to avoid breaches of minimum capital requirements.

30.2 A capital conservation buffer of 2.5%, comprised of Common Equity Tier 1 (CET1), is established above the regulatory minimum capital requirement. Capital distribution constraints will be imposed on a bank when capital levels fall within this range. Banks will be able to conduct business as normal when their capital levels fall into the conservation range as they experience losses. The constraints imposed only relate to distributions, not the operation of the bank.

Common Equity Tier 1 must first be used to meet the minimum capital and total loss-absorbing capacity (TLAC) requirements if necessary (including the 6% Tier 1, 8% Total capital requirements), before the remainder can contribute to the capital conservation buffer.

30.4 The table below shows the minimum capital conservation ratios (CET1 + CCoB) a bank must meet at various levels of CET1 capital ratios. The applicable conservation standards must be recalculated at each distribution date. For example, a bank with a CET1 capital ratio in the range of 5.125% to 5.75% is required to conserve 80% of its earnings in the subsequent payment period (ie pay out no more than 20% in terms of dividends, share buybacks and discretionary bonus payments). If the bank wants to make payments in excess of the constraints imposed by this regime, it would have the option of raising capital in the private sector equal to the amount above the constraint which it wishes to distribute. This would be discussed with the bank's supervisor as part of the capital planning process. The CET1 ratio includes amounts used to meet the

4.5% minimum CET1 requirement, but excludes any additional CET1 needed to meet the 6% Tier 1 and 8% Total Capital requirements, and also excludes any CET1 needed to meet the total lossabsorbing capacity (TLAC) requirement.

For example, a bank with 8% CET1 and no Additional Tier 1 or Tier 2 capital, that has 10% of non-regulatory-capital TLAC instruments, would meet its minimum risk-based capital and risk-based TLAC requirements, but would have a zero conservation buffer and therefore be subject to the 100% constraint on capital distributions.

Individual bank minimum capital conservation standards	
CET1 Ratio	Minimum Capital Conservation Ratios (expressed as a percentage of earnings)
4.5% - 5.125%	100%
>5.125% - 5.75%	80%
>5.75% - 6.375%	60%
>6.375% - 7.0%	40%
> 7.0%	0%

Countercyclical Buffer (CCyB)

30.6 Losses incurred in the banking sector can be extremely large when a downturn is preceded by a period of excess credit growth. These losses can destabilise the banking sector and spark a vicious circle, whereby problems in the financial system can contribute to a downturn in the real economy that then feeds back on to the banking sector. These interactions highlight the particular importance of the banking sector building up additional capital defences in periods where the risks of system-wide stress are growing markedly.

30.7 The countercyclical buffer aims to ensure that banking sector capital requirements take account of the macro-financial environment in which banks operate. It will be deployed by national jurisdictions when excess aggregate credit growth is judged to be associated with a build-up of system-wide risk to ensure the banking system has a buffer of capital to protect it against future potential losses. This focus on excess aggregate credit growth means that jurisdictions are likely to only need to deploy the buffer on an infrequent basis. The buffer for internationally-active banks will be a weighted average of the buffers deployed across all the jurisdictions to which it has credit exposures. This means that they will likely find themselves subject to a small buffer on a more frequent basis, since credit cycles are not always highly correlated across jurisdictions.

National CCyB Requirements

30.9 Each Basel Committee member jurisdiction will identify an authority with the responsibility to make decisions on the size of the countercyclical capital buffer. If the relevant national authority judges a period of excess credit growth to be leading to the build up of system-wide risk, they will consider, together with any other macroprudential tools at their disposal, putting in place a countercyclical buffer requirement. This will

vary between 0% and 2.5% of risk weighted assets, depending on their judgement as to the extent of the build up of system-wide risk.

30.11 To give banks time to adjust to a buffer level, a jurisdiction will pre-announce its decision to raise the level of the countercyclical buffer by up to 12 months.

30.18 For illustrative purposes, the following table sets out the conservation ratios a bank must meet at various levels of CET1 capital if the bank is subject to a 2.5% countercyclical buffer requirement.

Common Equity Tier 1 Ratio (including other fully loss absorbing capital)	Minimum Capital Conservation Ratios (expressed as a percentage of earnings)
4.5% - 5.75%	100%
>5.75% - 7.0%	80%
>7.0% - 8.25%	60%
>8.25% - 9.5%	40%
> 9.5%	0%

Capital Conservation Best Practice

30.20 Outside of periods of stress, banks should hold buffers of capital above the regulatory minimum. Retaining a greater proportion of earnings during a downturn will help ensure that capital remains available to support the ongoing business operations of banks through the period of stress.

30.21 When buffers have been drawn down, one way banks should look to rebuild them is through reducing discretionary distributions of earnings. This could include reducing dividend payments, share-backs and staff bonus payments. Banks may also choose to raise new capital from the private sector as an alternative to conserving internally generated capital

RBC40: SIB Buffers

This chapter describes the higher loss absorbency requirements applying to global and domestic systemically important banks.

Higher Loss Absorbency: G-SIBs

40.1 Taking into account this going-concern objective, global systemically important banks (G-SIBs) must meet their higher loss absorbency requirement with Common Equity Tier 1 capital only.

40.4 Based on the bucketing approach set out in SCO40.19 to SCO40.22, the magnitude of the higher loss absorbency requirement for each bucket is as follows.

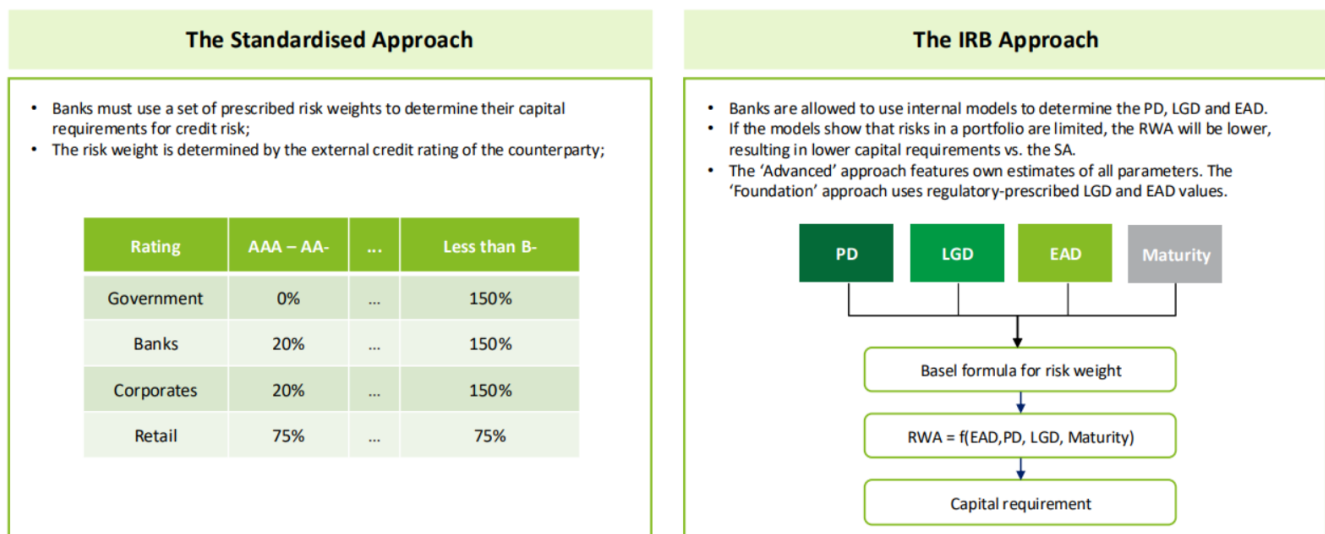
Bucket	Score range	Higher loss absorbency requirement (common equity as a percentage of risk-weighted assets)
5	530-629	3.5%
4	430-529	2.5%
3	330-429	2.0%
2	230-329	1.5%
1	130-229	1.0%

Higher Loss Absorbency: D-SIBs

40.7 As described in SCO50, a domestic systemically important bank (D-SIB) framework is best understood as taking the complementary perspective to the GSIB regime by focusing on the impact that the distress or failure of banks (including by international banks) will have on the domestic economy.

CRE

This standard describes how to calculate capital requirements for credit risk.



CRE20: Standardised Approach (Individual Exposures)

This chapter sets out the standardised approach for credit risk as it applies to individual claims.

Sovereigns

20.7 Exposures to sovereigns and their central banks will be risk-weighted as follows:

Risk weight table for sovereigns and central banks

Table 1

External rating	AAA to AA–	A+ to A–	BBB+ to BBB–	BB+ to B–	Below B–	Unrated
Risk weight	0%	20%	50%	100%	150%	100%

20.9 For the purpose of risk-weighting exposures to sovereigns, supervisors may recognise the country risk scores assigned by Export Credit Agencies (ECAs). These ECA risk scores will correspond to risk weight categories as detailed below.

Risk weight table for sovereigns and central banks

Table 2

ECA risk scores	0 to 1	2	3	4 to 6	7
Risk weight	0%	20%	50%	100%	150%

20.10 Exposures to the Bank for International Settlements, the International Monetary Fund, the European Central Bank, the European Union, the European Stability Mechanism and the European Financial Stability Facility may receive a 0% risk weight.

Public Sector Entities

20.11 Exposures to domestic PSEs will be risk-weighted at national discretion, according to either of the following two options.

Option 1: Based on external rating of sovereign

Table 3

External rating of the sovereign	AAA to AA–	A+ to A–	BBB+ to BBB–	BB+ to B–	Below B–	Unrated
Risk weight under Option 1	20%	50%	100%	100%	150%	100%

Risk weight table for PSEs

Option 2: Based on external rating of PSE

Table 4

External rating of the PSE	AAA to AA–	A+ to A–	BBB+ to BBB–	BB+ to B–	Below B–	Unrated
Risk weight under Option 2	20%	50%	50%	100%	150%	50%

Banks

20.16 For the purposes of calculating capital requirements, a bank exposure is defined as a claim (including loans and senior debt instruments, unless considered as subordinated debt for the purposes of CRE20.60) on any financial institution that is licensed to take deposits from the public and is subject to appropriate prudential standards and level of supervision

20.17 Bank exposures will be risk-weighted based on the following hierarchy:

1. External Credit Risk Assessment Approach (ECRA): This approach is for banks incorporated in jurisdictions that allow the use of external ratings for regulatory purposes. It applies to all their rated exposures to banks. Banks will apply CRE21.1 to CRE21.21 to determine which rating can be used and for which exposures.
2. Standardised Credit Risk Assessment Approach (SCRA): This approach is for all exposures of banks incorporated in jurisdictions that do not allow the use of external ratings for regulatory purposes. For exposures to banks that are unrated, this approach also applies to banks incorporated in jurisdictions that allow the use of external ratings for regulatory purposes.

ECRA

20.18 Banks incorporated in jurisdictions that allow the use of external ratings for regulatory purposes will assign to their rated bank exposures the corresponding “base” risk weights determined by the external ratings according to Table 6.

Risk weight table for bank exposures

External Credit Risk Assessment Approach

Table 6

External rating of counterparty	AAA to AA–	A+ to A–	BBB+ to BBB–	BB+ to B–	Below B–
“Base” risk weight	20%	30%	50%	100%	150%
Risk weight for short-term exposures	20%	20%	20%	50%	150%

SCRA

20.21 Banks incorporated in jurisdictions that do not allow the use of external ratings for regulatory purposes will apply the SCRA to all their bank exposures. The SCRA also applies to unrated bank exposures for banks incorporated in jurisdictions that allow the use of external ratings for regulatory purposes. The SCRA requires banks to classify bank exposures into one of three risk-weight buckets (ie Grades A, B and C) and assign the corresponding risk weights in Table 7.

Risk weight table for bank exposures

Standardised Credit Risk Assessment Approach

Table 7

Credit risk assessment of counterparty	Grade A	Grade B	Grade C
“Base” risk weight	40%	75%	150%
Risk weight for short-term exposures	20%	50%	150%

20.22 Grade A refers to exposures to banks, where the counterparty bank has adequate capacity to meet their financial commitments (including repayments of principal and interest) in a timely manner, for the projected life of the assets or exposures and irrespective of the economic cycles and business conditions.

20.25 Grade B refers to exposures to banks, where the counterparty bank is subject to substantial credit risk, such as repayment capacities that are dependent on stable or favourable economic or business conditions.

20.28 Grade C refers to higher credit risk exposures to banks, where the counterparty bank has material default risks and limited margins of safety. For these counterparties, adverse business, financial, or economic conditions are very likely to lead, or have led, to an inability to meet their financial commitments.

Securities Firms & Other Financial Institutions

20.40 Exposures to securities firms and other financial institutions will be treated as exposures to banks provided that these firms are subject to prudential standards and a level of supervision equivalent to those applied to banks (including capital and liquidity requirements)

Corporates

20.41 For the purposes of calculating capital requirements, exposures to corporates include exposures (loans, bonds, receivables, etc) to incorporated entities, associations, partnerships, proprietorships, trusts, funds and other entities with similar characteristics, except those which qualify for one of the other exposure classes. The treatment associated with subordinated debt and equities of these counterparties is addressed in CRE20.53 to CRE20.62. The corporate exposure class includes exposures to insurance companies and other financial corporates that do not meet the definitions of exposures to banks, or securities firms and other financial institutions, as determined in CRE20.16 and CRE20.40 respectively. The corporate exposure class does not include exposures to individuals. The corporate exposure class differentiates between the following subcategories:

- 1. General corporate exposures;
- 2. Specialised lending exposures, as defined in CRE20.48.

General Corporate Exposures

20.42 For corporate exposures of banks incorporated in jurisdictions that allow the use of external ratings for regulatory purposes, banks will assign "base" risk weights according to Table 10.

20.43 Unrated corporate exposures of banks incorporated in jurisdictions that allow the use of external ratings for regulatory purposes will receive a 100% risk weight, with the exception of unrated exposures to corporate small or medium-sized entities (SMEs), as described in CRE20.47.

Risk weight table for corporate exposures						
Jurisdictions that use external ratings for regulatory purposes						Table 10
External rating of counterparty	AAA to AA–	A+ to A–	BBB+ to BBB–	BB+ to BB–	Below BB–	Unrated
"Base" risk weight	20%	50%	75%	100%	150%	100%

20.44 For corporate exposures of banks incorporated in jurisdictions that do not allow the use of external ratings for regulatory purposes, banks will assign a 100% risk weight to all corporate exposures, with the exception of:

- 1. exposures to corporates identified as "investment grade" in CRE20.46; and
- 2. exposures to corporate SMEs in CRE20.47

20.46 Banks in jurisdictions that do not allow the use of external ratings for regulatory purposes may assign a 65% risk weight to exposures to "investment grade" corporates. An "investment grade" corporate is a

corporate entity that has adequate capacity to meet its financial commitments in a timely manner and its ability to do so is assessed to be robust against adverse changes in the economic cycle and business conditions.

SMEs

20.47 Corporate SMEs are defined as corporate exposures where the reported annual sales for the consolidated group of which the corporate counterparty is a part is less than or equal to €50 million for the most recent financial year. In some jurisdictions (eg emerging economies), national supervisors might deem it appropriate to define SMEs in a more conservative manner (ie with a lower level of sales). For unrated exposures to corporate SMEs in jurisdictions that allow the use of external ratings for regulatory purposes, and for all exposures to corporate SMEs in jurisdictions that do not allow the use of external ratings for regulatory purposes, an 85% risk weight will be applied. Exposures to SMEs that meet the criteria in CRE20.65(1) to CRE20.65(3) will be treated as regulatory retail SME exposures and risk weighted at 75%.

Specialised Lending

20.48 A corporate exposure will be treated as a specialised lending exposure if such lending possesses some or all of the following characteristics, either in legal form or economic substance:

- The exposure is not related to real estate and is within the definitions of object finance, project finance or commodities finance under CRE20.49. If the activity is related to real estate, the treatment would be determined in accordance with CRE20.69 to CRE20.91;

20.49 Exposures described in CRE20.48 will be classified in one of the following three subcategories of specialised lending:

1. Project finance refers to the method of funding in which the lender looks primarily to the revenues generated by a single project, both as the source of repayment and as security for the loan. This type of financing is usually for large, complex and expensive installations such as power plants, chemical processing plants, mines, transportation infrastructure, environment, media, and telecoms.
2. Object finance refers to the method of funding the acquisition of equipment (eg ships, aircraft, satellites, railcars, and fleets) where the repayment of the loan is dependent on the cash flows generated by the specific assets that have been financed and pledged or assigned to the lender.
3. Commodities finance refers to short-term lending to finance reserves, inventories, or receivables of exchange-traded commodities (eg crude oil, metals, or crops), where the loan will be repaid from the proceeds of the sale of the commodity and the borrower has no independent capacity to repay the loan.

20.50 Banks incorporated in jurisdictions that allow the use of external ratings for regulatory purposes will assign to their specialised lending exposures the risk weights determined by the issue-specific external ratings, if these are available, according to Table 10. Issuer ratings must not be used (ie CRE21.12 does not apply in the case of specialised lending exposures).

20.51 For specialised lending exposures for which an issue-specific external rating is not available, and for all specialised lending exposures of banks incorporated in jurisdictions that do not allow the use of external ratings for regulatory purposes, the following risk weights will apply:

- Object and commodities finance exposures will be risk-weighted at 100%;

- Project finance exposures will be risk-weighted at 130% during the preoperational phase and 100% during the operational phase. Project finance exposures in the operational phase which are deemed to be high quality, as described in CRE20.52, will be risk weighted at 80%. For this purpose, operational phase is defined as the phase in which the entity that was specifically created to finance the project has (a) a positive net cash flow that is sufficient to cover any remaining contractual obligation, and (b) declining long-term debt

20.52 A high quality project finance exposure refers to an exposure to a project finance entity that is able to meet its financial commitments in a timely manner and its ability to do so is assessed to be robust against adverse changes in the economic cycle and business conditions.

Equity, Subordinated Debt & Other Instruments

20.53 The treatment described in CRE20.57 to CRE20.60 applies to subordinated debt, equity and other regulatory capital instruments issued by either corporates or banks, provided that such instruments are not deducted from regulatory capital or risk-weighted at 250% according to CAP30, or risk weighted at 1250% according to CRE20.62. It also excludes equity investments in funds treated under CRE60.

20.54 Equity exposures are defined on the basis of the economic substance of the instrument. They include both direct and indirect ownership interests,²⁰ whether voting or non-voting, in the assets and income of a commercial enterprise or of a financial institution that is not consolidated or deducted. An instrument is considered to be an equity exposure if it meets all of the following requirements:

1. It is irredeemable in the sense that the return of invested funds can be achieved only by the sale of the investment or sale of the rights to the investment or by the liquidation of the issuer;
2. It does not embody an obligation on the part of the issuer; and
3. It conveys a residual claim on the assets or income of the issuer.

Indirect equity interests include holdings of derivative instruments tied to equity interests, and holdings in corporations, partnerships, limited liability companies or other types of enterprises that issue ownership interests and are engaged principally in the business of investing in equity instruments.

20.55 In addition to instruments classified as equity as a result of paragraph CRE20.54, the following instruments must be categorised as an equity exposure:

- An instrument with the same structure as those permitted as Tier 1 capital for banking organisations.

20.56 Debt obligations and other securities, partnerships, derivatives or other vehicles structured with the intent of conveying the economic substance of equity ownership are considered an equity holding. This includes liabilities from which the return is linked to that of equities.

20.57 Banks will assign a risk weight of 400% to speculative unlisted equity exposures described in CRE20.58 and a risk weight of 250% to all other equity holdings, with the exception of those equity holdings referred to in CRE20.59.

20.58 Speculative unlisted equity exposures are defined as equity investments in unlisted companies that are invested for short-term resale purposes or are considered venture capital or similar investments which are subject to price volatility and are acquired in anticipation of significant future capital gains.

20.60 Banks will assign a risk weight of 150% to subordinated debt and capital instruments other than equities. Any liabilities that meet the definition of "other TLAC liabilities" in CAP30.3 to CAP30.5 and that are not deducted from regulatory capital are considered to be subordinated debt for the purposes of this paragraph.

Retail Lending

20.63 The retail exposure class excludes exposures within the real estate exposure class. The retail exposure class includes the following types of exposures:

1. exposures to an individual person or persons; and
2. exposures to SMEs (as defined in CRE20.47) that meet the "regulatory retail" criteria set out in CRE20.65(1) to CRE20.65(3) below. Exposures to SMEs that do not meet these criteria will be treated as corporate SMEs exposures under CRE20.47.

20.64 For the purpose of determining risk weighted assets, the retail exposure class consists of the follow three sets of exposures:

1. "Regulatory retail" exposures that do not arise from exposures to "transactors" (as defined in CRE20.66).
2. "Regulatory retail" exposures to "transactors".
3. "Other retail" exposures.

20.65 "Regulatory retail" exposures are defined as retail exposures that meet all of the criteria listed below:

1. Product criterion: the exposure takes the form of any of the following: revolving credits and lines of credit (including credit cards, charge cards and overdrafts), personal term loans and leases (eg instalment loans, auto loans and leases, student and educational loans, personal finance) and small business facilities and commitments. Mortgage loans, derivatives and other securities (such as bonds and equities), whether listed or not, are specifically excluded from this category.
2. Low value of individual exposures: the maximum aggregated exposure to one counterparty cannot exceed an absolute threshold of €1 million.
3. Granularity criterion: no aggregated exposure to one counterparty can exceed 0.2% of the overall regulatory retail portfolio, unless national supervisors have determined another method to ensure satisfactory diversification of the regulatory retail portfolio. Defaulted retail exposures are to be excluded from the overall regulatory retail portfolio when assessing the granularity criterion.

20.66 "Transactors" are obligors in relation to facilities such as credit cards and charge cards where the balance has been repaid in full at each scheduled repayment date for the previous 12 months. Obligors in relation to overdraft facilities would also be considered as transactors if there has been no drawdown over the previous 12 months

20.67 "Other retail" exposures are defined as exposures to an individual person or persons that do not meet all of the regulatory retail criteria in CRE20.65.

20.68 The risk weights that apply to exposures in the retail asset class are as follows:

1. Regulatory retail exposures that do not arise from exposures to transactors (as defined in CRE20.66) will be risk weighted at 75%.
2. Regulatory retail exposures that arise from exposures to transactors (as defined in CRE20.66) will be risk weighted at 45%.

3. Other retail exposures will be risk weighted at 100%

Real Estate

20.69 Real estate is immovable property that is land, including agricultural land and forest, or anything treated as attached to land, in particular buildings, in contrast to being treated as movable/personal property. The real estate exposure asset class consists of:

1. Exposures secured by real estate that are classified as "regulatory real estate" exposures.
2. Exposures secured by real estate that are classified as "other real estate" exposures.
3. Exposures that are classified as "land acquisition, development and construction" (ADC) exposures.

20.70 "Regulatory real estate" exposures consist of:

1. "Regulatory residential real estate" exposures that are not "materially dependent on cash flows generated by the property".
2. "Regulatory residential real estate" exposures that are "materially dependent on cash flows generated by the property".
3. "Regulatory commercial real estate" exposures that are not "materially dependent on cash flows generated by the property".
4. "Regulatory commercial real estate" exposures that are "materially dependent on cash flows generated by the property".

20.71 For an exposure secured by real estate to be classified as a "regulatory real estate" exposure, the loan must meet the following requirements:

- Finished property: the exposure must be secured by a fully completed immovable property. Subject to national discretion, supervisors may allow this criteria to be met by loans to individuals that are secured by residential property under construction or land upon which residential property would be constructed, provided that: (i) the property is a one-to-four family residential housing unit that will be the primary residence of the borrower and the lending to the individual is not, in effect, indirectly financing land acquisition, development and construction exposures described in CRE20.90; or (ii) sovereign or PSEs involved have the legal powers and ability to ensure that the property under construction will be finished.
- Ability of the borrower to repay: the borrower must meet the requirements set according to CRE20.73.
- Prudent value of property: the property must be valued according to the criteria in CRE20.74 to CRE20.76 for determining the value in the loan-to-value ratio (LTV). Moreover, the value of the property must not depend materially on the performance of the borrower.

LTV

20.74 The LTV is the amount of the loan divided by the value of the property. When calculating the LTV, the loan amount will be reduced as the loan amortises. The value of the property will be maintained at the value measured at origination, with the following exceptions:

1. The national supervisors elect to require banks to revise the property value downward. If the value has been adjusted downwards, a subsequent upwards adjustment can be made but not to a higher value than the value at origination.

2. The value must be adjusted if an extraordinary, idiosyncratic event occurs resulting in a permanent reduction of the property value.
3. Modifications made to the property that unequivocally increase its value could also be considered in the LTV.

20.75 The LTV must be prudently calculated in accordance with the following requirements:

1. Amount of the loan: includes the outstanding loan amount and any undrawn committed amount of the mortgage loan.
2. Value of the property: the valuation must be appraised independently using prudently conservative valuation criteria

Residential Real Estate

20.77 A “regulatory residential real estate” exposure is a regulatory real estate exposure that is secured by a property that has the nature of a dwelling and satisfies all applicable laws and regulations enabling the property to be occupied for housing purposes (ie residential property)

20.82 For regulatory residential real estate exposures that are not materially dependent on cash flow generated by the property, the risk weight to be assigned to the total exposure amount will be determined based on the exposure’s LTV ratio in Table 11 below. The use of the risk weights in Table 11 is referred to as the “whole loan” approach.

Whole loan approach risk weights for regulatory residential real estate exposures that are not materially dependent on cash flows generated by the property

Table 11

	LTV ≤ 50%	50% < LTV ≤ 60%	60% < LTV ≤ 80%	80% < LTV ≤ 90%	90% < LTV ≤ 100%	LTV > 100%
Risk weight	20%	25%	30%	40%	50%	70%

20.83 As an alternative to the whole loan approach for regulatory residential real estate exposures that are not materially dependent on cash flows generated by the property, jurisdictions may apply the “loan splitting” approach. Under the loan splitting approach, the risk weight of 20% is applied to the part of the exposure up to 55% of the property value and the risk weight of the counterparty (as prescribed in CRE20.89(1)) is applied to the residual exposure.

For example, for a loan of €70,000 to an individual secured on a property valued at €100,000, the bank will apply a risk weight of 20% to €55,000 of the exposure and, according to CRE20.89(1), a risk weight of 75% to the residual exposure of €15,000. This gives total risk weighted assets for the exposure of €22,250 = (0.20 €55,000) + (0.75 €15,000).

20.84 For regulatory residential real estate exposures that are materially dependent on cash flows generated by the property, the risk weight to be assigned to the total exposure amount will be determined based on the

exposure's LTV ratio in Table 12 below.

Risk weights for regulatory residential real estate exposures that are materially dependent on cash flows generated by the property						Table 12
	LTV ≤ 50%	50% < LTV ≤ 60%	60% < LTV ≤ 80%	80% < LTV ≤ 90%	90% < LTV ≤ 100%	LTV > 100%
Risk weight	30%	35%	45%	60%	75%	105%

Commercial Real Estate

20.78 A "regulatory commercial real estate" exposure is regulatory real estate exposure that is not a regulatory residential real estate exposure.

20.85 For regulatory commercial real estate exposures that are not materially dependent on cash flow generated by the property, the risk weight to be assigned to the total exposure amount will be determined based on the exposure's LTV in Table 13 below (which sets out a whole loan approach). The risk weight of the counterparty for the purposes of Table 13 below and CRE20.86 below is prescribed in CRE20.89(1).

Whole loan approach risk weights for regulatory commercial real estate exposures that are not materially dependent on cash flows generated by the property			Table 13
	LTV ≤ 60%	LTV > 60%	
Risk weight	Min (60%, RW of counterparty)	RW of counterparty	

20.86 As an alternative to the whole loan approach for regulatory commercial real estate exposures that are not materially dependent on cash flows generated by the property, jurisdictions may apply the "loan splitting" approach. Under the loan splitting approach, the risk weight of 60% or the risk weight of the counterparty, whichever is lower, is applied to the part of the exposure up to 55% of the property value³⁸, and the risk weight of the counterparty is applied to the residual exposure.

20.87 For regulatory commercial real estate exposures that are materially dependent on cash flows generated by the property³⁹, the risk weight to be assigned to the total exposure amount will be determined based on the exposure's LTV in Table 14 below.

Whole loan approach risk weights for regulatory commercial real estate exposures that are materially dependent on cash flows generated by the property			Table 14
	LTV $\leq$ 60%	60% < LTV $\leq$ 80%	LTV > 80%
Risk weight	70%	90%	110%

Other Real Estate

20.88 An “other real estate” exposure is an exposure within the real estate asset class that is not a regulatory real estate exposure (as defined in CRE20.71 above) and is not a land ADC exposure (as defined in CRE20.90 below).

20.89 Other real estate exposures are risk weighted as follows:

- The risk weight of the counterparty is used for other real estate exposures that are not materially dependent on the cash flows generated by the property. For exposures to individuals the risk weight applied will be 75%. For exposures to SMEs, the risk weight applied will be 85%. For exposures to other counterparties, the risk weight applied is the risk weight that would be assigned to an unsecured exposure to that counterparty.
- The risk weight of 150% is used for other real estate exposures that are materially dependent on the cash flows generated by the property

Land ADC

20.90 Land ADC exposures refers to loans to companies or SPVs financing any of the land acquisition for development and construction purposes, or development and construction of any residential or commercial property. ADC exposures will be risk-weighted at 150%, unless they meet the criteria in CRE20.91.

ADC exposures do not include the acquisition of forest or agricultural land, where there is no planning consent or intention to apply for planning consent.

20.91 ADC exposures to residential real estate may be risk weighted at 100%, provided that the following criteria are met: 20.91 (1) prudential underwriting standards meet the requirements in CRE20.71 (ie the requirements that are used to classify regulatory real estate exposures) where applicable; (2) pre-sale or pre-lease contracts amount to a significant portion of total contracts or substantial equity at risk. Pre-sale or pre-lease contracts must be legally binding written contracts and the purchaser/renter must have made a substantial cash deposit which is subject to forfeiture if the contract is terminated. Equity at risk should be determined as an appropriate amount of borrower-contributed equity to the real estate’s appraised as-completed value.

Derivatives

20.102 For exposures that give rise to counterparty credit risk according to CRE51.4 (ie OTC derivatives, exchange-traded derivatives, long settlement transactions and securities financing transactions), the exposure amount to be used in the determination of RWA is to be calculated under the rules set out in CRE50 to CRE54.

Credit Derivatives

20.103 A bank providing credit protection through a first-to-default or second-to-default credit derivative is subject to capital requirements on such instruments. For first-to-default credit derivatives, the risk weights of the assets included in the basket must be aggregated up to a maximum of 1250% and multiplied by the nominal amount of the protection provided by the credit derivative to obtain the risk-weighted asset amount.

Defaulted Exposures (Stage 3)

20.104 For risk-weighting purposes under the standardised approach, a defaulted exposure is defined as one that is past due for more than 90 days, or is an exposure to a defaulted borrower.

20.105 For retail exposures, the definition of default can be applied at the level of a particular credit obligation, rather than at the level of the borrower. As such, default by a borrower on one obligation does not require a bank to treat all other obligations to the banking group as defaulted.

20.106 With the exception of residential real estate exposures treated under CRE20.107, the unsecured or unguaranteed portion of a defaulted exposure shall be risk-weighted net of specific provisions and partial write-offs as follows:

1. 150% risk weight when specific provisions are less than 20% of the outstanding amount of the loan; and
2. 100% risk weight when specific provisions are equal or greater than 20% of the outstanding amount of the loan.

20.107 Defaulted residential real estate exposures where repayments do not materially depend on cash flows generated by the property securing the loan shall be riskweighted net of specific provisions and partial write-offs at 100%. Guarantees or financial collateral which are eligible according to the credit risk mitigation framework might be taken into account in the calculation of the exposure in accordance with CRE20.76.

CRE30: IRB Approach Overview

This chapter sets out an overview of the internal ratings-based approach to credit risk, including the categorisation of exposures, a description of the available approaches and the roll-out requirements.

30.1 This chapter describes the internal ratings-based (IRB) approach for credit risk. Subject to certain minimum conditions and disclosure requirements, banks that have received supervisory approval to use the IRB approach may rely on their own internal estimates of risk components in determining the capital requirement for a given exposure. The risk components include measures of the probability of default (PD), loss given default (LGD), the exposure at default (EAD), and effective maturity (M). In some cases, banks may be required to use a supervisory value as opposed to an internal estimate for one or more of the risk components.

30.2 The IRB approach is based on measures of unexpected losses (UL) and expected losses. The risk-weight functions, as outlined in CRE31, produce capital requirements for the UL portion. Expected losses are treated separately, as outlined in CRE35.

30.4 Under the IRB approach, banks must categorise banking-book exposures into broad classes of assets with different underlying risk characteristics, subject to the definitions set out below. The classes of assets are (a) corporate, (b) sovereign, (c) bank, (d) retail, and (e) equity. Within the corporate asset class, five sub-classes

of specialised lending are separately identified. Within the retail asset class, three sub-classes are separately identified. Within the corporate and retail asset classes, a distinct treatment for purchased receivables may also apply provided certain conditions are met. For the equity asset class the IRB approach is not permitted, as outlined further below.

Corporate Class

30.6 In general, a corporate exposure is defined as a debt obligation of a corporation, partnership, or proprietorship. Banks are permitted to distinguish separately exposures to small or medium-sized entities (SME), as defined in CRE31.8.

30.7 In addition to general corporates, within the corporate asset class five sub-classes of specialised lending (SL) are identified.

30.8 The five sub-classes of SL are project finance (PF), object finance (OF), commodities finance (CF), income-producing real estate (IPRE) lending, and highvolatility commercial real estate (HVCRE) lending. Each of these sub-classes is defined below.

30.9 PF is a method of funding in which the lender looks primarily to the revenues generated by a single project, both as the source of repayment and as security for the exposure

30.11 OF refers to a method of funding the acquisition of physical assets (eg ships, aircraft, satellites, railcars, or fleets) where the repayment of the exposure is dependent on the cash flows generated by the specific assets that have been financed and pledged or assigned to the lender.

30.12 CF refers to structured short-term lending to finance reserves, inventories, or receivables of exchange-traded commodities (eg crude oil, metals, or crops), where the exposure will be repaid from the proceeds of the sale of the commodity and the borrower has no independent capacity to repay the exposure.

30.14 IPRE lending refers to a method of providing funding to real estate (such as, office buildings to let, retail space, multifamily residential buildings, industrial or warehouse space, or hotels) where the prospects for repayment and recovery on the exposure depend primarily on the cash flows generated by the asset. The primary source of these cash flows would generally be lease or rental payments or the sale of the asset.

30.15 HVCRE lending is the financing of commercial real estate that exhibits higher loss rate volatility (ie higher asset correlation) compared to other types of SL. HVCRE includes:

1. Commercial real estate exposures secured by properties of types that are categorised by the national supervisor as sharing higher volatilities in portfolio default rates;
2. Loans financing any of the land acquisition, development and construction (ADC) phases for properties of those types in such jurisdictions; and

Sovereign Class

30.17 This asset class covers all exposures to counterparties treated as sovereigns under the standardised approach. This includes sovereigns (and their central banks), certain public sector entities (PSEs) identified as sovereigns in the standardised approach, multilateral development banks (MDBs) that meet the criteria for a 0% risk weight and referred to in the first footnote of CRE20.14 of the standardised approach, and the entities referred to in CRE20.10 of the standardised approach.

Bank Class

30.18 This asset class covers exposures to banks as defined in CRE20.16 of the standardised approach for credit risk and those securities firms and other financial institutions set out in CRE20.40 of the standardised approach for credit risk that are treated as exposures to banks. This asset class also includes exposures to the entities listed in this paragraph that are in the form of subordinated debt or regulatory capital instruments (which form their own asset class within the standardised approach), provided that such instruments: (i) do not fall within the scope of equity exposures as defined in CRE30.26; (ii) are not deducted from regulatory capital or risk-weighted at 250% according to CAP30; and (iii) are not risk weighted at 1250% according to CRE20.62.

Retail Class

30.19 An exposure is categorised as a retail exposure if it meets all of the criteria set out in CRE30.20 (which relate to the nature of the borrower and value of individual exposures) and all of the criteria set out in CRE30.22 (which relate to the size of the pool of exposures).

30.20 The criteria related to the nature of the borrower and value of the individual exposures are as follows:

1. Exposures to individuals – such as revolving credits and lines of credit (eg credit cards, overdrafts, or retail facilities secured by financial instruments) as well as personal term loans and leases (eg instalment loans, auto loans and leases, student and educational loans, personal finance, or other exposures with similar characteristics) – are generally eligible for retail treatment regardless of exposure size, although supervisors may wish to establish exposure thresholds to distinguish between retail and corporate exposures.
2. Where a loan is a residential mortgage (including first and subsequent liens, term loans and revolving home equity lines of credit) it is eligible for retail treatment regardless of exposure size so long as the credit is: (a) an exposure to an individual; or (b) an exposure to associations or cooperatives of individuals that are regulated under national law and exist with the only purpose of granting its members the use of a primary residence in the property securing the loan.
3. Where loans are extended to small businesses and managed as retail exposures they are eligible for retail treatment provided the total exposure of the banking group to a small business borrower (on a consolidated basis where applicable) is less than €1 million. Small business loans extended through or guaranteed by an individual are subject to the same exposure threshold.

30.23 Within the retail asset class category, banks are required to identify separately three sub-classes of exposures:

1. residential mortgage loans, as defined above;
2. qualifying revolving retail exposures, as defined in the following paragraph; and
3. all other retail exposures.

30.24 All of the following criteria must be satisfied for a sub-portfolio to be treated as a qualifying revolving retail exposure (QRRE). These criteria must be applied at a sub-portfolio level consistent with the bank's segmentation of its retail activities generally. Segmentation at the national or country level (or below) should be the general rule.

1. The exposures are revolving, unsecured, and uncommitted (both contractually and in practice). In this context, revolving exposures are defined as those where customers' outstanding balances are permitted

- to fluctuate based on their decisions to borrow and repay, up to a limit established by the bank.
- 2. The exposures are to individuals.
- 3. The maximum exposure to a single individual in the sub-portfolio is €100,000 or less.

30.25 The QRRE sub-class is split into exposures to transactors and revolvers. A QRRE transactor is an exposure to an obligor that meets the definition set out in CRE20.64 of the standardised approach. That is, the exposure is to an obligor in relation to a facility such as credit card or charge card where the balance has been repaid in full at each scheduled repayment date for the previous 12 months, or the exposure is in relation to an overdraft facility if there have been no drawdowns over the previous 12 months. All exposures that are not QRRE transactors are QRRE revolvers, including QRRE exposures with less than 12 months of repayment history.

Equity Class

30.26 This asset class covers exposures to equities as defined in CRE20.54 to CRE20.56 of the standardised approach for credit risk

F-IRB vs A-IRB

30.32 For each of the asset classes covered under the IRB framework, there are three key elements:

1. Risk components: estimates of risk parameters provided by banks, some of which are supervisory estimates.
2. Risk-weight functions: the means by which risk components are transformed into risk-weighted assets and therefore capital requirements.
3. Minimum requirements: the minimum standards that must be met in order for a bank to use the IRB approach for a given asset class.

30.33 For certain asset classes, the Committee has made available two broad approaches: a foundation and an advanced approach.

- Under the foundation approach (F-IRB approach), as a general rule, banks provide their own estimates of PD and rely on supervisory estimates for other risk components.
- Under the advanced approach (A-IRB approach), banks provide their own estimates of PD, LGD and EAD, and their own calculation of M, subject to meeting minimum standards.
- For both the foundation and advanced approaches, banks must always use the risk-weight functions provided in this Framework for the purpose of deriving capital requirements. The full suite of approaches is described below.

30.34 For exposures to equities, as defined in CRE30.26 above, the IRB approaches are not permitted (see CRE30.43). In addition, the A-IRB approach cannot be used for the following:

1. Exposures to general corporates (ie exposures to corporates that are not classified as specialised lending) belonging to a group with total consolidated annual revenues greater than €500m.
2. Exposures in the bank asset class CRE30.18, and other securities firms and financial institutions (including insurance companies and any other financial institutions in the corporate asset class).

30.35 In making the assessment for the revenue threshold in CRE30.34 above, the amounts must be as reported in the audited financial statements of the corporates or, for corporates that are part of consolidated

groups, their consolidated groups (according to the accounting standard applicable to the ultimate parent of the consolidated group). The figures must be based on the average amounts calculated over the prior three years, or on the latest amounts updated every three years by the bank.

Corporate, Sovereign & Bank Exposures

30.36 Under the foundation approach, banks must provide their own estimates of PD associated with each of their borrower grades, but must use supervisory estimates for the other relevant risk components. The other risk components are LGD, EAD and M.

30.37 Under the advanced approach, banks must calculate the effective maturity (M) and provide their own estimates of PD, LGD and EAD.

30.38 There is an exception to this general rule for the five sub-classes of assets identified as SL.

Specilised Lending & Commercial Real Estate

30.39 Banks that do not meet the requirements for the estimation of PD under the corporate foundation approach for their SL exposures are required to map their internal risk grades to five supervisory categories, each of which is associated with a specific risk weight. This version is termed the 'supervisory slotting criteria approach'.

30.40 Banks that meet the requirements for the estimation of PD are able to use the foundation approach to corporate exposures to derive risk weights for all classes of SL exposures except HVCRE. At national discretion, banks meeting these requirements for HVCRE exposures are able to use a foundation approach that is similar in all respects to the corporate approach, with the exception of a separate risk-weight function as described in CRE31.11.

30.41 Banks that meet the requirements for the estimation of PD, LGD and EAD are able to use the advanced approach to corporate exposures to derive risk weights for all classes of SL exposures except HVCRE. At national discretion, banks meeting these requirements for HVCRE exposure are able to use an advanced approach that is similar in all respects to the corporate approach, with the exception of a separate risk-weight function as described in CRE31.11.

Retail Exposure

30.42 For retail exposures, banks must provide their own estimates of PD, LGD and EAD. There is no foundation approach for this asset class.

Roll-out Requirements

30.45. Once a bank adopts an IRB approach for part of its holdings within an asset class, it is expected to extend it across all holdings within that asset class. In this context, the relevant assets classes are as follows:

1. Sovereigns
2. Banks
3. Corporates (excluding specialised lending and purchased receivables)
4. Specialised lending
5. Corporate purchased receivables

6. QRRE
7. Retail residential mortgages
8. Other retail (excluding purchased receivables)
9. Retail purchased receivables

30.46 The Committee recognises that, for many banks, it may not be practicable for various reasons to implement the IRB approach for an entire asset class across all business units at the same time. Furthermore, once on IRB, data limitations may mean that banks can meet the standards for the use of own estimates of LGD and EAD for some but not all of their exposures within an asset class at the same time (for example, exposures that are in the same asset class, but are in different business units).

30.47 As such, supervisors may allow banks to adopt a phased rollout of the IRB approach across an asset class. The phased rollout includes:

1. adoption of IRB across the asset class within the same business unit;
2. adoption of IRB for the asset class across business units in the same banking group; and
3. move from the foundation approach to the advanced approach for certain risk components where use of the advanced approach is permitted.

However, when a bank adopts an IRB approach for an asset class within a particular business unit, it must apply the IRB approach to all exposures within that asset class in that unit.

30.48 If a bank intends to adopt an IRB approach to an asset class, it must produce an implementation plan, specifying to what extent and when it intends to roll out the IRB approaches within the asset class and business units. The plan should be realistic, and must be agreed with the supervisor. It should be driven by the practicality and feasibility of moving to the more advanced approaches, and not motivated by a desire to adopt an approach that minimises its capital charge. During the roll-out period, supervisors will ensure that no capital relief is granted for intra-group transactions which are designed to reduce a banking group's aggregate capital charge by transferring credit risk among entities on the standardised approach, foundation and advanced IRB approaches. This includes, but is not limited to, asset sales or cross guarantees.

30.49 Some exposures that are immaterial in terms of size and perceived risk profile within their asset class may be exempt from the requirements in the previous two paragraphs, subject to supervisory approval. Capital requirements for such operations will be determined according to the standardised approach, with the national supervisor determining whether a bank should hold more capital under the supervisory review process standard (SRP) for such positions.

30.50 Banks adopting an IRB approach for an asset class are expected to continue to employ an IRB approach for that asset class. A voluntary return to the standardised or foundation approach is permitted only in extraordinary circumstances, such as divestiture of a large fraction of the bank's credit-related business in that asset class, and must be approved by the supervisor

30.51 Given the data limitations associated with SL exposures, a bank may remain on the supervisory slotting criteria approach for one or more of the PF, OF, CF, IPRE or HVCRE sub-classes, and move to the foundation or advanced approach for the other sub-classes. However, a bank should not move to the advanced approach for the HVCRE sub-class without also doing so for material IPRE exposures at the same time

Equity

30.43 All equity exposures are subject to the approach set out in CRE20.57 of the standardised approach for credit risk, with the exception of equity investments in funds that are subject to the requirements set out in CRE60.

CRE31: IRB Approach Risk-weight Functions

31.1 This chapter presents the calculation of risk weighted assets under the internal ratings-based (IRB) approach for: (i) corporate, sovereign and bank exposures; and (ii) retail exposures. Risk weighted assets are designed to address unexpected losses from exposures. The method of calculating expected losses, and for determining the difference between that measure and provisions, is described CRE35

31.2 Regarding the risk-weight functions for deriving risk weighted assets set out in this chapter:

- Probability of default (PD) and loss-given-default (LGD) are measured as decimals
- Exposure at default (EAD) is measured as currency (eg euros), except where explicitly noted otherwise
- $\ln$ denotes the natural logarithm
- $N(x)$ denotes the cumulative distribution function for a standard normal random variable (ie the probability that a normal random variable with mean zero and variance of one is less than or equal to x). The normal cumulative distribution function is, for example, available in Excel as the function **NORMSDIST**.
- $G(z)$ denotes the inverse cumulative distribution function for a standard normal random variable (ie the value of x such that $N(x) = z$). The inverse of the normal cumulative distribution function is, for example, available in Excel as the function **NORMSINV**.

```
from scipy.stats import norm

x = 1.3
N_x = norm.cdf(x)

z = 0.95
G_z = norm.ppf(z)

print(N_x)
print(G_z)
```

Defaulted Exposures (ST3)

31.3 The capital requirement K for a defaulted exposure is equal to the greater of zero and the difference between its LGD (described in CRE36.83) and the bank's best estimate of expected loss (described in CRE36.86). The risk-weighted asset amount for the defaulted exposure is the product of K , 12.5, and the EAD.

$$RWA = \max\{0, 12.5 \times (LGD - EL_{BE}) \times EAD\}$$

Corporates, Sovereigns & Banks

31.4 The derivation of risk-weighted assets is dependent on estimates of the PD, LGD, EAD and, in some cases, effective maturity (M), for a given exposure.

31.5 For exposures not in default, the formula for calculating risk-weighted assets is as follows (illustrative risk weights are shown in CRE99):

$$\text{Correlation } R = 0.12 \left(\frac{1 - e^{-50PD}}{1 - e^{-50}} \right) + 0.24 \left(1 - \frac{1 - e^{-50PD}}{1 - e^{-50}} \right)$$

$$\text{Maturity Adjustment } b = [0.11852 - 0.05478 \ln(\text{PD})]^2$$

$$\text{Capital Req } K = [N \left(\frac{G(PD) + \sqrt{R}}{\sqrt{1-R}} \right) \times G(0.999) \times \text{LGD} - PD \times \text{LGD}] \times \frac{1 + (M - 2.5)b}{1 - 1.5b}$$

$$\text{RWA } = 12.5 \times K \times \text{EAD}$$

31.6 Regarding the formula set out in CRE31.5 above, M is the effective maturity, calculated according to CRE32.43 to CRE32.54 and the following term is used to refer to a specific part of the capital requirements formula:

$$\text{Full Maturity Adjustment } = \frac{1 + (M - 2.5)b}{1 - 1.5b}$$

31.7 A multiplier of 1.25 is applied to the correlation parameter of all exposures to financial institutions meeting the following criteria:

1. Regulated financial institutions whose total assets are greater than or equal to USD100 billion.
2. Unregulated financial institutions, regardless of size

$$\text{Correlation } R_{FI} = 1.25 R$$

SME

31.8 Under the IRB approach for corporate credits, banks will be permitted to separately distinguish exposures to SME borrowers (defined as corporate exposures where the reported sales for the consolidated group of which the firm is a part is less than €50 million) from those to large firms. A firm-size adjustment is made to the corporate risk weight formula for exposures to SME borrowers. S is expressed as total annual sales in millions of euros with values of S falling in the range of equal to or less than €50 million or greater than or equal to €5 million. Reported sales of less than €5 million will be treated as if they were equivalent to €5 million for the purposes of the firm-size adjustment for SME borrowers.

$$\text{Correlation } R = R - 0.04 \left(\frac{S - 5}{45} \right)$$

Specialised Lending

31.10 Regarding project finance, object finance, commodities finance and income producing real estate sub-asset classes of specialised lending (SL):

1. Banks that meet the requirements for the estimation of PD will be able to use the foundation IRB (F-IRB) approach for the corporate asset class to derive risk weights for SL sub-classes. As specified in CRE33.2, banks that do not meet the requirements for the estimation of PD will be required to use the supervisory slotting approach.
2. Banks that meet the requirements for the estimation of PD, LGD and EAD (where relevant) will be able to use the advanced IRB (A-IRB) approach for the corporate asset class to derive risk weights for SL sub-classes.

31.11 Regarding the high volatility commercial real estate (HVCRE) sub-asset class of specialised lending, banks that meet the requirements for the estimation of PD and whose supervisor has chosen to implement a foundation or advanced approach to HVCRE exposures will use the same formula for the derivation of risk weights that is used for other SL exposures, except that they will apply the following asset correlation formula:

$$\text{Correlation } R = 0.12 \left(\frac{1 - e^{-50PD}}{1 - e^{-50}} \right) + 0.30 \left(1 - \frac{1 - e^{-50PD}}{1 - e^{-50}} \right)$$

31.12 Banks that do not meet the requirements for estimation of LGD or EAD for HVCRE exposures must use the supervisory parameters for LGD and EAD for corporate exposures, or use the supervisory slotting approach.

Retail Lending

31.13 There are three separate risk-weight functions for retail exposures, as defined in CRE31.14 to CRE31.16. Risk weights for retail exposures are based on separate assessments of PD and LGD as inputs to the risk-weight functions. None of the three retail risk-weight functions contain the full maturity adjustment component that is present in the risk-weight function for exposures to banks, sovereigns and corporates. Illustrative risk weights are shown in CRE99.

Retail Residential Mortgage

31.14 For exposures defined in CRE30.19 that are not in default and are secured or partly secured by residential mortgages, risk weights will be assigned based on the following formula:

$$\text{Correlation } R = 0.15$$

$$\text{Capital Req } K = \text{LGD} \left[N \left(\frac{G(PD) + \sqrt{R} \times G(0.999)}{\sqrt{1-R}} \right) - PD \right]$$

$$\text{RWA } = 12.5 \times K \times \text{EAD}$$

QRRE

31.15 For qualifying revolving retail exposures as defined in CRE30.23 and CRE30.24 that are not in default, risk weights are defined based on the following formula:

$$\text{Correlation } R = 0.04$$

$$\text{Capital Req } K = \text{LGD} \left[N \left(\frac{G(PD) + \sqrt{R} \times G(0.999)}{\sqrt{1-R}} \right) - PD \right]$$

$$\text{RWA } = 12.5 \times K \times \text{EAD}$$

Other Retail Exposures

31.16 For all other retail exposures that are not in default, risk weights are assigned based on the following function, which allows correlation to vary with PD:

$$\text{Correlation } R = 0.03 \left(\frac{1 - e^{-50PD}}{1 - e^{-50}} \right) + 0.16 \left(1 - \frac{1 - e^{-50PD}}{1 - e^{-50}} \right)$$

$$\text{Capital Req } K = \text{LGD} \left[N \left(\frac{G(PD) + \sqrt{R} \times G(0.999)}{\sqrt{1-R}} \right) - PD \right]$$

$$\text{RWA } = 12.5 \times K \times \text{EAD}$$

CRE32: IRB Approach Risk Components

32.1 This chapter presents the calculation of the risk components (PD, LGD, EAD, M) that are used in the formulas set out in CRE31. In calculating these components, the legal certainty standards for recognising credit risk mitigation under the standardised approach to credit risk (CRE22) apply for both the foundation and advanced internal ratings-based (IRB) approaches.

Corporate, Sovereign and Banks

32.2 This section, CRE32.2 to CRE32.56, sets out the calculation of the risk components for corporate, sovereign and bank exposures.

PDs

32.3 For corporate, sovereign and bank exposures, the PD is the one-year PD associated with the internal borrower grade to which that exposure is assigned. The PD of borrowers assigned to a default grade(s), consistent with the reference definition of default, is 100%. The minimum requirements for the derivation of the PD estimates associated with each internal borrower grade are outlined in CRE36.77 to CRE36.79.

32.4 With the exception of exposures in the sovereign asset class, the PD for each exposure that is used as input into the risk weight formula and the calculation of expected loss must not be less than 0.05%.

LGD: F-IRB

32.6 Under the foundation approach, senior claims on sovereigns, banks, securities firms and other financial institutions (including insurance companies and any financial institutions in the corporate asset class) that are not secured by recognised collateral will be assigned a 45% LGD. Senior claims on other corporates that are not secured by recognised collateral will be assigned a 40% LGD.

32.7 All subordinated claims on corporates, sovereigns and banks will be assigned a 75% LGD. A subordinated loan is a facility that is expressly subordinated to another facility. At national discretion, supervisors may choose to employ a wider definition of subordination. This might include economic subordination, such as cases where the facility is unsecured and the bulk of the borrower's assets are used to secure other exposures.

32.8 to 32.14 give a in-depth description of how to deal with collateral

LGD: A-IRB

32.15 Subject to certain additional minimum requirements specified below (and the conditions set out in CRE30.34), supervisors may permit banks to use their own internal estimates of LGD for corporate and sovereign exposures. LGD must be measured as the loss given default as a percentage of the EAD. Banks eligible for the IRB approach that are unable to meet these additional minimum requirements must utilise the foundation LGD treatment described above.

32.16 The LGD for each corporate exposure that is used as input into the risk weight formula and the calculation of expected loss must not be less than the parameter floors indicated in the table below (the floors do not apply to the LGD for exposures in the sovereign asset class):

LGD parameter floors for corporate exposures	
Unsecured	Secured
25%	Varying by collateral type: • 0% financial • 10% receivables • 10% commercial or residential real estate • 15% other physical

32.17 to 32.18 discuss how to account for collateralised exposures.

32.19 The minimum requirements for the derivation of LGD estimates are outlined in CRE36.83 to CRE36.88.

Exposure at default (EAD)

32.29 The following sections apply to both on and off-balance sheet positions. All exposures are measured gross of specific provisions or partial write-offs. The EAD on drawn amounts should not be less than the sum of: (i) the amount by which a bank's regulatory capital would be reduced if the exposure were written-off fully; and (ii) any specific provisions and partial write-offs. When the difference between the instrument's EAD and the sum of (i) and (ii) is positive, this amount is termed a discount. The calculation of risk-weighted assets is independent of any discounts. Under the limited circumstances described in CRE35.4, discounts may be included in the measurement of total eligible provisions for purposes of the EL-provision calculation set out in CRE35.

A credit card with a £100,000 credit limit has £20,000 drawn down. The bank has a specific provision of £2,000 for potential losses. If the entire £20,000 were written off, the bank's regulatory capital would reduce by £20,000. The minimum EAD on the drawn amount is £20,000 (i) + £2,000 (ii) = £22,000.

32.30 On-balance sheet netting of loans and deposits will be recognised subject to the same conditions as under CRE22.68 of the standardised approach.

32.31 For off-balance sheet items there are two approaches for the estimation of EAD: a foundation approach and an advanced approach. When only the drawn balances of revolving facilities have been securitised, banks must ensure that they continue to hold required capital against the undrawn balances associated with the securitised exposures.

32.32 In the foundation approach, EAD is calculated as the committed but undrawn amount multiplied by a credit conversion factor (CCF). In the advanced approach, EAD for undrawn commitments may be calculated as the committed but undrawn amount multiplied by a CCF or derived from direct estimates of total facility EAD. In both the foundation approach and advanced approaches, the definition of commitments is the same as in the standardised approach, as set out in CRE20. 94.

EAD: F-IRB

32.33 The types of instruments and the CCFs applied to them under the F-IRB approach are the same as those in the standardised approach, as set out in CRE20.94 to CRE20.101

32.34 The amount to which the CCF is applied is the lower of the value of the unused committed credit line, and the value that reflects any possible constraining of the availability of the facility, such as the existence of a ceiling on the potential lending amount which is related to a borrower's reported cash flow.

EAD: A-IBR

32.36 Banks which meet the minimum requirements for use of their own estimates of EAD (see CRE36.89 to CRE36.98) will be allowed for exposures for which A-IRB is permitted (see CRE30.33) to use their own internal estimates of EAD for undrawn revolving commitments to extend credit, purchase assets or issue credit substitutes provided the exposure is not subject to a CCF of 100% in the foundation approach (see CRE32.33). Standardised approach CCFs must be used for all other off-balance sheet items (for example, undrawn non-revolving commitments), and must be used where the minimum requirements for own estimates of EAD are not met. The EAD for each exposure that is not in the sovereign asset class that is used as input into the risk weight formula and the calculation of expected loss is subject to a floor that is the sum of: (i) the on balance sheet amount; and (ii) 50% of the off balance sheet exposure using the applicable CCF in the standardised approach.

A revolving loan facility is one that lets a borrower obtain a loan where the borrower has the flexibility to decide how often to withdraw from the loan and at what time intervals. A revolving facility allows the borrower to drawdown, repay and re-draw loans advanced to it. Facilities that allow prepayments and subsequent redraws of those prepayments are considered as revolving.

Effective Maturity (M)

32.44 Effective maturity (M) will be 2.5 years for exposures to which the bank applies the foundation approach, except for repo-style transactions where the effective maturity is 6 months (ie $M=0.5$). National supervisors may choose to require all banks in their jurisdiction (those using the foundation and advanced approaches) to measure M for each facility using the definition provided below

32.45 Banks using any element of the A-IRB approach are required to measure effective maturity for each facility as defined below. However, national supervisors may allow the effective maturity to be fixed at 2.5 years (the "fixed maturity treatment") for facilities to certain smaller domestic corporate borrowers if the reported sales (ie turnover) as well as total assets for the consolidated group of which the firm is a part of are less than €500 million.

32.46 Except as noted in CRE32.51, the effective maturity (M) is subject to a floor of one year and a cap of 5 years.

32.47 For an instrument subject to a determined cash flow schedule, effective maturity M is defined as follows, where CF_t denotes the cash flows (principal, interest, payments and fees) contractually payable by the borrower in period t

$$\text{Effective Maturity } M = \frac{\sum_t t \cdot CF_t}{\sum_t CF_t}$$

32.48 If a bank is not in a position to calculate the effective maturity of the contracted payments as noted above, it is allowed to use a more conservative measure of M such as that it equals the maximum remaining

time (in years) that the borrower is permitted to take to fully discharge its contractual obligation (principal, interest, and fees) under the terms of loan agreement. Normally, this will correspond to the nominal maturity of the instrument

Retail

32.57 This section, CRE32.57 to CRE32.67, sets out the calculation of the risk components for retail exposures.

PDs & LGDs

32.58 For each identified pool of retail exposures, banks are expected to provide an estimate of the PD and LGD associated with the pool, subject to the minimum requirements as set out in CRE36. Additionally, the PD for retail exposures is the greater of:

- (i) the one-year PD associated with the internal borrower grade to which the pool of retail exposures is assigned; and
- (ii) 0.1% for qualifying revolving retail exposure (QRRE) revolvers (see CRE30.24 for the definition of QRRE revolvers) and 0.05% for all other exposures.

The LGD for each exposure that is used as input into the risk weight formula and the calculation of expected loss must not be less than the parameter floors indicated in the table below:

LGD parameter floors for retail exposures		
Type of exposure	Unsecured	Secured
Mortgages	Not applicable	5%
QRRE (transactors and revolvers)	50%	Not applicable
Other retail	30%	Varying by collateral type: • 0% financial • 10% receivables • 10% commercial or residential real estate • 15% other physical

32.59 Regarding the LGD parameter floors set out in the table above, the LGD floors for partially secured exposures in the “other retail” category should be calculated according to the formula set out in CRE32.17. The LGD floor for residential mortgages is fixed at 5%, irrespective of the level of collateral provided by the property.

EAD

32.62 Both on- and off-balance sheet retail exposures are measured gross of specific provisions or partial write-offs. The EAD on drawn amounts should not be less than the sum of: (i) the amount by which a bank's regulatory capital would be reduced if the exposure were written-off fully; and (ii) any specific provisions and partial write-offs. When the difference between the instrument's EAD and the sum of (i) and (ii) is positive, this amount is termed a discount. The calculation of risk-weighted assets is independent of any discounts. Under the limited circumstances described in CRE35.4, discounts may be included in the measurement of total eligible provisions for purposes of the EL-provision calculation set out in chapter CRE35.

32.63 Banks must use their own estimates of EAD for undrawn revolving commitments to extend credit, purchase assets or issue credit substitutes provided the exposure is not subject to a CCF of 100% in the standardised approach (see CRE20.92) and the minimum requirements in CRE36.89 to CRE36.99 are satisfied.

32.65 For retail exposures with uncertain future drawdown such as credit cards, banks must take into account their history and/or expectation of additional drawings prior to default in their overall calibration of loss estimates. In particular, where a bank does not reflect conversion factors for undrawn lines in its EAD estimates, it must reflect in its LGD estimates the likelihood of additional drawings prior to default. Conversely, if the bank does not incorporate the possibility of additional drawings in its LGD estimates, it must do so in its EAD estimates.

32.66 When only the drawn balances of revolving retail facilities have been securitised, banks must ensure that they continue to hold required capital against the undrawn balances associated with the securitised exposures using the IRB approach to credit risk for commitments

CRE35: IRB Approach ELs and Provisions

35.1 This chapter discusses the calculation of expected losses (EL) under the internal ratings-based (IRB) approach, and the method by which the difference between provisions (eg specific provisions, partial write-offs, portfolio-specific general provisions such as country risk provisions or general provisions) and EL may be included in or must be deducted from regulatory capital, as outlined in the definition of capital standard (CAP10.19 and CAP30.13). The treatment of EL and provisions related to securitisation exposures is outlined in CRE40.36.

Expected Losses

35.2 A bank must sum the EL amount (defined as EL multiplied by exposure at default) associated with its exposures to which the IRB approach is applied (excluding the EL amount associated with securitisation exposures) to obtain a total EL amount.

35.3 Banks must calculate EL as probability of default (PD) times loss-given-default (LGD) for corporate, sovereign, bank, and retail exposures not in default. For corporate, sovereign, bank, and retail exposures that are in default, banks must use their best estimate of expected loss as defined in CRE36.86 for exposures subject to the advanced approach and for exposures subject to the foundation approach banks must use the supervisory LGD. For exposures subject to the supervisory slotting criteria EL is calculated as described in the chapter on the supervisory slotting approach (paragraphs CRE33.8 to CRE33.12). Securitisation exposures do not contribute to the EL amount, as set out in CRE40.36.

Provisions

35.4 Total eligible provisions are defined as the sum of all provisions (eg specific provisions, partial write-offs, portfolio-specific general provisions such as country risk provisions or general provisions) that are attributed to exposures treated under the IRB approach. In addition, total eligible provisions may include any discounts on defaulted assets. General and specific provisions set aside against securitisation exposures must not be included in total eligible provisions.

35.8 As specified in CAP10.19 and CAP30.13, banks using the IRB approach must compare the total amount of total eligible provisions (as defined in paragraph CRE35.4) with the total EL amount as calculated within the IRB approach (as defined in paragraph CRE35.2). In addition, CAP10.18 outlines the treatment for that portion of a bank that is subject to the standardised approach for credit risk when the bank uses both the standardised and IRB approaches.

35.9 Where the calculated EL amount is lower than the total eligible provisions of the bank, its supervisors must consider whether the EL fully reflects the conditions in the market in which it operates before allowing the difference to be included in Tier 2 capital. If specific provisions exceed the EL amount on defaulted assets this assessment also needs to be made before using the difference to offset the EL amount on non-defaulted assets.

CRE36: IRB Approach Minimum Requirements

36.1 This chapter presents the minimum requirements for entry and on-going use of the internal ratings-based (IRB) approach. The minimum requirements are set out in the following 11 sections:

1. Composition of minimum requirements
2. Compliance with minimum requirements
3. Rating system design
4. Risk rating system operations
5. Corporate governance and oversight
6. Use of internal ratings
7. Risk quantification
8. Validation of internal estimates
9. Supervisory loss-given-default (LGD) and exposure at default (EAD) estimates
10. Requirements for recognition of leasing
11. Disclosure requirements

Section 3: Rating System Design

36.9 The term “rating system” comprises all of the methods, processes, controls, and data collection and IT systems that support the assessment of credit risk, the assignment of internal risk ratings, and the quantification of default and loss estimates.

36.10 Within each asset class, a bank may utilise multiple rating methodologies /systems.

Corporate, Sovereign and Banks

36.11 A qualifying IRB rating system must have two separate and distinct dimensions:

- the risk of borrower default; and
- transaction-specific factors.

36.12 The first dimension must be oriented to the risk of borrower default. Separate exposures to the same borrower must be assigned to the same borrower grade, irrespective of any differences in the nature of each specific transaction.

36.13 The second dimension must reflect transaction-specific factors, such as collateral, seniority, product type, etc.

Retail Exposures

36.16 Rating systems for retail exposures must be oriented to both borrower and transaction risk, and must capture all relevant borrower and transaction characteristics. Banks must assign each exposure that falls within the definition of retail for IRB purposes into a particular pool. Banks must demonstrate that this process provides for a meaningful differentiation of risk, provides for a grouping of sufficiently homogenous exposures, and allows for accurate and consistent estimation of loss characteristics at pool level.

36.17 For each pool, banks must estimate PD, LGD, and EAD. Multiple pools may share identical PD, LGD and EAD estimates. At a minimum, banks should consider the following risk drivers when assigning exposures to a pool:

- Borrower risk characteristics (eg borrower type, demographics such as age /occupation).
- Transaction risk characteristics, including product and/or collateral types (eg loan to value measures, seasoning, guarantees; and seniority (first vs. second lien)). Banks must explicitly address crosscollateral provisions where present.
- Delinquency of exposure: Banks are expected to separately identify exposures that are delinquent and those that are not.

Section 7: Risk Quantification

Overall Requirements

36.62 This section addresses the broad standards for own-estimates of PD, LGD, and EAD. Generally, all banks using the IRB approaches must estimate a PD for each internal borrower grade for corporate, sovereign and bank exposures or for each pool in the case of retail exposures.

36.63 PD estimates must be a long-run average of one-year default rates for borrowers in the grade, with the exception of retail exposures as set out in CRE36.81 and CRE36.82. Requirements specific to PD estimation are provided in CRE36.77 to CRE36.82. Banks on the advanced approach must estimate an appropriate LGD (as defined in CRE36.83 to CRE36.88) for each of its facilities (or retail pools). For exposures subject to the advanced approach, banks must also estimate an appropriate long-run default-weighted average EAD for each of its facilities as defined in CRE36.89 and CRE36.90. Requirements specific to EAD estimation appear in CRE36.89 to CRE36.99. For corporate, sovereign and bank exposures, banks that do not meet the requirements for own-estimates of EAD or LGD, above, must use the supervisory estimates of these parameters. Standards for use of such estimates are set out in CRE36.128 to CRE36.145

36.66 The population of exposures represented in the data used for estimation, and lending standards in use when the data were generated, and other relevant characteristics should be closely matched to or at least comparable with those of the bank's exposures and standards. The bank must also demonstrate that economic or market conditions that underlie the data are relevant to current and foreseeable conditions. For estimates of LGD and EAD, banks must take into account CRE36.83 to CRE36.99. The number of exposures in

the sample and the data period used for quantification must be sufficient to provide the bank with confidence in the accuracy and robustness of its estimates. The estimation technique must perform well in out-of-sample tests.

Margin of Conservatism (MoC)

36.67 In general, estimates of PDs, LGDs, and EADs are likely to involve unpredictable errors. In order to avoid over-optimism, a bank must add to its estimates a margin of conservatism that is related to the likely range of errors. Where methods and data are less satisfactory and the likely range of errors is larger, the margin of conservatism must be larger.

Appropriate Adjustment (AA)

Supervisors may allow some flexibility in application of the required standards for data that are collected prior to the date of implementation of this Framework. However, in such cases banks must demonstrate to their supervisors that appropriate adjustments have been made to achieve broad equivalence to the data without such flexibility. Data collected beyond the date of implementation must conform to the minimum standards unless otherwise stated.

Default Definition

36.68 A default is considered to have occurred with regard to a particular obligor when either or both of the two following events have taken place.

1. The bank considers that the obligor is unlikely to pay its credit obligations to the banking group in full, without recourse by the bank to actions such as realising security (if held).
2. The obligor is past due more than 90 days on any material credit obligation to the banking group. Overdrafts will be considered as being past due once the customer has breached an advised limit or been advised of a limit smaller than current outstandings.

36.71 For retail exposures, the definition of default can be applied at the level of a particular facility, rather than at the level of the obligor. As such, default by a borrower on one obligation does not require a bank to treat all other obligations to the banking group as defaulted.

36.72 A bank must record actual defaults on IRB exposure classes using this reference definition. A bank must also use the reference definition for its estimation of PDs, and (where relevant) LGDs and EADs.

Loss Definition

36.76 The definition of loss used in estimating LGD is economic loss. When measuring economic loss, all relevant factors should be taken into account. This must include material discount effects and material direct and indirect costs associated with collecting on the exposure. Banks must not simply measure the loss recorded in accounting records, although they must be able to compare accounting and economic losses. The bank's own workout and collection expertise significantly influences their recovery rates and must be reflected in their LGD estimates, but adjustments to estimates for such expertise must be conservative until the bank has sufficient internal empirical evidence of the impact of its expertise.

PD Requirements: Corporates, Sovereigns and Banks

36.78 For all methods listed below, banks must estimate a PD for each rating grade based on the observed historical average one-year default rate that is a simple average based on number of obligors (count weighted). Weighting approaches, such as EAD weighting, are not permitted.

1. A bank may use data on internal default experience for the estimation of PD. A bank must demonstrate in its analysis that the estimates are reflective of underwriting standards and of any differences in the rating system that generated the data and the current rating system. Where only limited data are available, or where underwriting standards or rating systems have changed, the bank must add a greater margin of conservatism in its estimate of PD. The use of pooled data across institutions may also be recognised. A bank must demonstrate that the internal rating systems and criteria of other banks in the pool are comparable with its own.
2. Banks may associate or map their internal grades to the scale used by an external credit assessment institution or similar institution and then attribute the default rate observed for the external institution's grades to the bank's grades. Mappings must be based on a comparison of internal rating criteria to the criteria used by the external institution and on a comparison of the internal and external ratings of any common borrowers. Biases or inconsistencies in the mapping approach or underlying data must be avoided. The external institution's criteria underlying the data used for quantification must be oriented to the risk of the borrower and not reflect transaction characteristics. The bank's analysis must include a comparison of the default definitions used, subject to the requirements in CRE36.68 to CRE36.73. The bank must document the basis for the mapping.
3. A bank is allowed to use a simple average of default-probability estimates for individual borrowers in a given grade, where such estimates are drawn from statistical default prediction models. The bank's use of default probability models for this purpose must meet the standards specified in CRE36.33

36.79 Irrespective of whether a bank is using external, internal, or pooled data sources, or a combination of the three, for its PD estimation, the length of the underlying historical observation period used must be at least five years for at least one source. If the available observation period spans a longer period for any source, and this data are relevant and material, this longer period must be used. The data should include a representative mix of good and bad years.

PD Requirements: Retail

36.80 Given the bank-specific basis of assigning exposures to pools, banks must regard internal data as the primary source of information for estimating loss characteristics. Banks are permitted to use external data or statistical models for quantification provided a strong link can be demonstrated between:

- (a) the bank's process of assigning exposures to a pool and the process used by the external data source; and
- (b) between the bank's internal risk profile and the composition of the external data. In all cases banks must use all relevant and material data sources as points of comparison.

36.81 One method for deriving long-run average estimates of PD and default-weighted average loss rates given default (as defined in CRE36.83) for retail would be based on an estimate of the expected long-run loss rate. A bank may (i) use an appropriate PD estimate to infer the long-run default-weighted average loss rate given default, or (ii) use a long-run default-weighted average loss rate given default to infer the appropriate PD. In either case, it is important to recognise that the LGD used for the IRB capital calculation cannot be less than the long-run default-weighted average loss rate given default and must be consistent with the concepts defined in CRE36.83.

36.82 Irrespective of whether banks are using external, internal, pooled data sources, or a combination of the three, for their estimation of loss characteristics, the length of the underlying historical observation period used must be at least five years. If the available observation spans a longer period for any source, and these data are relevant, this longer period must be used. The data should include a representative mix of good and bad years of the economic cycle relevant for the portfolio. The PD should be based on the observed historical average one-year default rate.

LGD Requirements

36.83 A bank must estimate an LGD for each facility that aims to reflect economic downturn conditions where necessary to capture the relevant risks. This LGD cannot be less than the long-run default-weighted average loss rate given default calculated based on the average economic loss of all observed defaults within the data source for that type of facility. In addition, a bank must take into account the potential for the LGD of the facility to be higher than the default-weighted average during a period when credit losses are substantially higher than average. For certain types of exposures, loss severities may not exhibit such cyclical variability and LGD estimates may not differ materially from the long-run default weighted average. However, for other exposures, this cyclical variability in loss severities may be important and banks will need to incorporate it into their LGD estimates. For this purpose, banks may make reference to the averages of loss severities observed during periods of high credit losses, forecasts based on appropriately conservative assumptions, or other similar methods. Appropriate estimates of LGD during periods of high credit losses might be formed using either internal and/or external data. Supervisors will continue to monitor and encourage the development of appropriate approaches to this issue.

36.85 LGD estimates must be grounded in historical recovery rates and, when applicable, must not solely be based on the collateral's estimated market value. This requirement recognises the potential inability of banks to gain both control of their collateral and liquidate it expeditiously. To the extent that LGD estimates take into account the existence of collateral, banks must establish internal requirements for collateral management, operational procedures, legal certainty and risk management process that are generally consistent with those required for the foundation IRB approach.

36.86 Recognising the principle that realised losses can at times systematically exceed expected levels, the LGD assigned to a defaulted asset should reflect the possibility that the bank would have to recognise additional, unexpected losses during the recovery period. For each defaulted asset, the bank must also construct its best estimate of the expected loss on that asset based on current economic circumstances and facility status. The amount, if any, by which the LGD on a defaulted asset exceeds the bank's best estimate of expected loss on the asset represents the capital requirement for that asset, and should be set by the bank on a risk-sensitive basis in accordance with CRE31.3. Instances where the best estimate of expected loss on a defaulted asset is less than the sum of specific provisions and partial charge-offs on that asset will attract supervisory scrutiny and must be justified by the bank.

36.87 Estimates of LGD must be based on a minimum data observation period that should ideally cover at least one complete economic cycle but must in any case be no shorter than a period of seven years for at least one source. If the available observation period spans a longer period for any source, and the data are relevant, this longer period must be used.

36.88 The minimum data observation period for LGD estimates for retail exposures is five years. The less data a bank has, the more conservative it must be in its estimation.

EAD Requirements

36.89 EAD for an on-balance sheet or off-balance sheet item is defined as the expected gross exposure of the facility upon default of the obligor.

36.90 Under the advanced approach, banks must assign an estimate of EAD for each eligible facility. It must be an estimate of the long-run default-weighted average EAD for similar facilities and borrowers over a sufficiently long period of time, but with a margin of conservatism appropriate to the likely range of errors in the estimate. If a positive correlation can reasonably be expected between the default frequency and the magnitude of EAD, the EAD estimate must incorporate a larger margin of conservatism. Moreover, for exposures for which EAD estimates are volatile over the economic cycle, the bank must use EAD estimates that are appropriate for an economic downturn, if these are more conservative than the long-run average. For banks that have been able to develop their own EAD models, this could be achieved by considering the cyclical nature, if any, of the drivers of such models. Other banks may have sufficient internal data to examine the impact of previous recession(s). However, some banks may only have the option of making conservative use of external data. Moreover, where a bank bases its estimates on alternative measures of central tendency (such as the median or a higher percentile estimate) or only on 'downturn' data, it should explicitly confirm that the basic downturn requirement of the framework is met, ie the bank's estimates do not fall below a (conservative) estimate of the long-run default-weighted average EAD for similar facilities.

36.93 Banks' EAD estimates must be developed using a 12-month fixed-horizon approach; ie for each observation in the reference data set, default outcomes must be linked to relevant obligor and facility characteristics twelve months prior to default.

36.94 Consistent with this principle, EAD estimates applied to particular exposures should not be based on data that conflate the effects of disparate characteristics or data from exposures that exhibit different characteristics (eg same broad product grouping but different customers that are managed differently by the bank). The estimates should be based on appropriately homogenous segments. Alternatively, the estimates should be based on an estimation approach that effectively disentangles the impact of the different characteristics exhibited within the relevant dataset. Practices that generally do not comply with this principle include use of estimates based or partly based on:

- (1) SME/midmarket data being applied to large corporate obligors.
- (2) Data from commitments with 'small' unused limit availability being applied to facilities with 'large' unused limit availability.
- (3) Data from obligors already identified as problematic at reference date being applied to current obligors with no known issues (eg customers at reference date who were already delinquent, watchlisted by the bank, subject to recent bank-initiated limit reductions, blocked from further drawdowns or subject to other types of collections activity).
- (4) Data that has been affected by changes in obligors' mix of borrowing and other credit-related products over the observation period unless that data has been effectively mitigated for such changes, eg by adjusting the data to remove the effects of the changes in the product mix

36.95 A well-known feature of the commonly used undrawn limit factor (ULF) approach to estimating CCFs is the region of instability associated with facilities close to being fully drawn at reference date. Banks should ensure that their EAD estimates are effectively quarantined from the potential effects of this region of instability.

- (1) An acceptable approach could include using an estimation method other than the ULF approach that avoids the instability issue by not using potentially small undrawn limits that could approach zero in the denominator or, as appropriate, switching to a method other than the ULF as the region of instability is approached, eg a limit factor, balance factor or additional utilisation factor approach. Note that, consistent with CRE36.94, including limit utilisation as a driver in EAD models could quarantine much of the relevant portfolio from this issue but, in the absence of other actions, leaves open how to develop appropriate EAD estimates to be applied to exposures within the region of instability.
- (2) Common but ineffective approaches to mitigating this issue include capping and flooring reference data (eg observed CCFs at 100 per cent and zero respectively) or omitting observations that are judged to be affected.

Undrawn Limit Factor Method

A specific type of CCF, where predicted additional drawings in the leadup to default are expressed as a percentage of the undrawn limit that remains available to the obligor under the terms and conditions of a facility

$$\text{EAD} = B_0 = B_t + \text{ULF}(L_t - B_t)$$
 where

- B_0 is the facility balance at date of default;
- B_t is the current balance (for predicted EAD) or balance at reference date (for observed EAD);
- L_t is the current limit (for predicted EAD) or limit at reference date (for realised/observed EAD).

Limit Factor Method

A limit factor (LF) is a specific type of CCF, where the predicted balance at default is expressed as a percentage of the total limit that is available to the obligor under the terms and conditions of a credit facility,

$$\text{EAD} = B_0 = \text{LF} \times L_t$$
 where

Balance Factor Method

A balance factor (BF) is a specific type of CCF, where the predicted balance at default is expressed as a percentage of the current balance that has been drawn down under a credit facility.

$$\text{EAD} = B_0 = \text{BF} \times B_t$$
 where

Additional Utilisation Factor

An additional utilisation factor (AUF) is a specific type of CCF, where predicted additional drawings in the lead-up to default are expressed as a percentage of the total limit that is available to the obligor under the terms and conditions of a credit facility.

$$\text{EAD} = B_0 = B_t + \text{AUF}(L_t)$$
 where

36.96 EAD reference data must not be capped to the principal amount outstanding or facility limits. Accrued interest, other due payments and limit excesses should be included in EAD reference data.

36.98 Estimates of EAD must be based on a time period that must ideally cover a complete economic cycle but must in any case be no shorter than a period of seven years. If the available observation period spans a

longer period for any source, and the data are relevant, this longer period must be used. EAD estimates must be calculated using a default-weighted average and not a time-weighted average.

36.99 The minimum data observation period for EAD estimates for retail exposures is five years. The less data a bank has, the more conservative it must be in its estimation.